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The Role of the Lumbardhi River in Sustainable Tourism Development in Peja: Integration of Ecological, Recreational, and Tourism Values

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Abstract: The Lumbardhi River connects the mountainous Rugova region with the urban area of Peja, providing a basis for integrating environmental protection with recreation and tourism planning. This exploratory study uses a purposive narrative synthesis of published secondary data and scientific literature on water quality, anthropogenic pressures, flood risk, river restoration, ecosystem services, recreation, and community participation. The synthesis identifies management priorities for pollution control, riparian protection, flood-sensitive access, and local participation. It proposes a preliminary framework comprising urban recreation, transition, and nature-oriented areas. These categories are planning concepts, not verified spatial zones or estimates of ecological carrying capacity. The evidence includes historical water-quality observations and studies conducted at different spatial scales; it does not establish current conditions throughout the corridor or demonstrate that any specific segment can accommodate increased use without ecological harm. Applying the framework therefore requires updated water-quality monitoring, field assessments of riparian habitats and riverbanks, spatial analysis of flood exposure, and evaluation of recreational impacts and community priorities. The framework provides a basis for further assessment and adaptive planning that considers environmental quality, recreational opportunities, and local benefits together. The framework combines river-specific environmental evidence, catchment-scale flood research, and considerations of recreation and local governance to identify what should be assessed before tourism use is expanded along the Peja–Rugova corridor.

Keywords: Lumbardhi River; Sustainable tourism; Water quality; Environmental management; Recreation; Ecosystem services

1. Introduction

The Lumbardhi River is a key feature of Peja's natural landscape and urban environment. It connects the mountainous Rugova region with the city, forming a corridor in which ecological, social, and recreational functions interact. In addition to its hydrological role, the river offers opportunities for recreation, improvements to urban public space, and nature-based tourism. It can therefore be considered part of Peja's green and blue infrastructure. A systematic review by Sousa et al. (2025) identifies a wide range of ecosystem services provided by urban rivers, including regulating and cultural services.

Local studies document historical variation in water quality (Faiku et al., 2015) and examine potential pressures from industry and food services (Dinaj, 2021). Their findings are limited to the periods and locations studied. Recreational opportunities also depend on spatial organization. Durán Vian et al. (2021) showed that the location of public spaces and access to water influence recreational interaction between a city and its river. For Peja, the river's continuity toward Rugova offers opportunities to connect urban recreation with nature-based activities. However, public access must be compatible with the corridor's ecological, morphological, and hydrological characteristics. River-restoration research indicates that restoration can enhance some ecosystem services and that integrated design should consider environmental, social, and economic dimensions (Guimarães et al., 2025; Kaiser

et al., 2020). Relevant measures include pollution control, protection of riparian vegetation, selective rehabilitation of degraded banks, and carefully planned access. These measures are consistent with research identifying natural resources and landscape quality as important components of Peja's tourism potential (Kuqi & Vishaj, 2024). Flood risk must also inform planning because urbanization influences flood extent in the Lumbardhi i Pejës catchment (Agaj et al., 2024).

From a community-development perspective, the corridor's tourism value depends on who participates in planning, who receives economic benefits, and how local knowledge informs management. Community-based tourism research cautions against treating host communities as homogeneous or assuming that participation automatically leads to equitable outcomes (Blackstock, 2005). Studies of tourism governance and value chains indicate that collaboration, local social networks, and community influence can shape access to employment, entrepreneurship, and development benefits (Chambwe & Saayman, 2025; Mtapuri et al., 2022). Integrated assessment should therefore consider ownership, employment, governance, livelihoods, and the distribution of benefits and costs alongside environmental and recreational outcomes (Simpson, 2009).

The river could link urban recreation in Peja with nature-based tourism toward Rugova and create opportunities for local enterprises (Kuqi & Vishaj, 2024; Mtapuri et al., 2022).

Such opportunities must be assessed alongside environmental limits. Evidence from residents' perceptions suggests that the economic, sociocultural, and environmental dimensions of tourism sustainability can evolve differently across stages of destination development (Lee & Jan, 2019). Improved access may enhance recreation while increasing pressure on riverbanks, waste-management services, or sensitive habitats. Protection measures should also consider everyday access and local economic participation. The central planning challenge is to identify forms and intensities of use compatible with the ecological and hydrological characteristics of individual river segments. This study therefore considers environmental quality, recreational opportunity, local benefit, and risk management as interdependent planning concerns.

Existing studies describe tourism resources and development potential in Peja and Rugova (Kuqi & Vishaj, 2024; Lajçi et al., 2022). This study combines historical water-quality and pollution evidence with catchment-scale flood research, recreation literature, and local governance considerations. Its contribution is to identify the conditions that should be assessed before recreation or tourism use is expanded along the river. Section 3.4 clarifies how this preliminary assessment framework differs from the existing Cityförster design concept.

2. Methodology

This study used an exploratory, purposive narrative synthesis of published scientific literature and secondary sources concerning the Lumbardhi i Pejës corridor, its urban section, and its continuity toward Rugova. Sources were selected for their relevance to environmental quality, anthropogenic pressures, recreation and tourism, hydrological risk, or community participation. River-specific studies provided local environmental evidence (Dinaj, 2021; Faiku et al., 2015), catchment-scale research informed the discussion of flood risk (Agaj et al., 2024), and studies of Peja and Rugova supplied destination context. Broader literature informed the interpretation of restoration, ecosystem services, recreation, and tourism governance.

Findings were extracted narratively by recording what each source reported, its observation or publication period where available, its spatial coverage, and the limitation relevant to corridor planning. The sources were compared across four analytical dimensions: environmental quality, anthropogenic pressures, recreational and tourism functions, and hydrological risk. Local benefits and governance were considered across all four dimensions. Management priorities were derived only where the combined evidence indicated a recurring planning concern; incomplete or divergent evidence was retained as uncertainty rather than resolved by assumption. Thus, historical measurements are used to justify repeat monitoring, catchment-scale flood findings to justify flood-sensitive assessment, and general literature to support principles rather than local effects. This procedure is interpretive and non-exhaustive, and the absence of a reproducible original search protocol remains a limitation.

The source set was assembled purposively around the questions addressed in the manuscript rather than through a protocol-driven systematic search. The original compilation combined known Lumbardhi-specific studies with published work on Peja/Rugova tourism and broader literature on urban rivers, restoration, recreation, and community-based tourism. A contemporaneous search log was not maintained; consequently, exact databases, search strings, and collection dates for the original source set cannot be reconstructed and are not claimed here. The Cityförster project page was consulted to establish the scope of the local river-corridor concept discussed in Section 3.4. Sources were retained when they contributed directly to one or more analytical dimensions or clarified local planning/governance context; sources were not treated as equivalent when their spatial scale, date, or evidentiary status differed.

For the principal sources, Table 1 records the observation period, where available, or the publication period when observation dates were not specified, together with spatial coverage, evidence used, limitations, and planning implications. Historical sampling dates were distinguished from publication dates. Catchment-scale and destination-level findings were not treated as direct measurements of individual river segments, and general

theoretical literature was distinguished from local observations.

The evidence was organized into four analytical dimensions: (1) environmental quality, (2) anthropogenic pressures, (3) recreational and tourism functions, and (4) hydrological risk. Local benefits and governance were considered across all four dimensions. Comparison of the sources informed provisional priorities for monitoring, pollution control, riparian protection, flood-sensitive access, and participation, as well as three preliminary planning categories. These categories represent the authors' interpretations and require further assessment. No new environmental measurements, systematic field survey, or quantitative comparison of river segments was undertaken. The synthesis does not establish current conditions throughout the corridor, verified zone boundaries, ecological carrying capacity, or the effects of increased recreational use.

2.1 Evidence Synthesis and Planning Implications

Table 1 summarizes the principal evidence used in the synthesis. The studies differ in date, purpose, and spatial scale; their planning implications reflect the authors' interpretations and do not constitute direct measurements or verified spatial boundaries.

Table 1. Local evidence and implications for corridor planning

Source	Period	Spatial Coverage	Evidence Used in This Study	Limitations and Planning Implications
Faiku et al. (2015)	Sampling: June 2013	Four sites: Haxhaj, "The city", "Transit bridge", and "Near the bridge of Klina"	Reported pH: 8.20–8.58; conductivity: 216–399 µS cm^{-1}; nitrate: 0.01–1.10 mg dm^{-3}; nitrite: 0.015–0.045 mg dm^{-3}	Historical, site-specific observations; neither current conditions nor a complete assessment of the Peja–Rugova corridor. Supports repeat monitoring before tourism decisions.
Dinaj (2021)	Publication: 2021	Four study areas along the Lumbardhi i Pejës	Describes a one-month investigation of physicochemical, macropollutant, and microbiological parameters in four areas, considering industrial and food-service pressures	Used to frame pollution-source assessment and monitoring priorities; no site-specific quantitative effects or current corridor-wide conditions are inferred here.
Agaj et al. (2024)	Publication: 2024	Lumbardhi i Pejës and Klina catchments	Uncontrolled urbanization was associated with higher peak discharge and larger flood-affected areas. Flood extent and spatial patterns were correlated with barriers, distance from natural flow paths, and slope	Catchment-scale flood-risk evidence; does not validate tourism-zone boundaries. Supports access and infrastructure planning informed by flood risk.
Kuqi (2025); Kuqi & Vishaj (2024); Lajçi et al. (2022)	Publications: 2022–2025	Peja and Rugova (coverage varies among studies)	Provide destination-level context on sustainable tourism potential, mountain tourism, and cultural heritage	Broader destination evidence, not direct measurement of corridor use or ecological capacity. Informs connections with nearby natural and cultural assets.
Durán Vian et al. (2021); Guimarães et al. (2025); Kaiser et al. (2020); Sousa et al. (2025)	Publications: 2020–2025	International research on urban rivers, restoration, recreation, and ecosystem services	Provide concepts for river–city interaction, ecosystem services, and integrated restoration planning	General literature rather than local evidence for Peja. Informs interpretation of principles without establishing measured local effects.
Blackstock (2005); Choi & Murray (2010); Choi & Sirakaya (2005)	Publications: 2005–2010	General literature on community-based tourism and residents' attitudes	Informs consideration of meaningful participation and community dimensions of sustainable tourism	Not specific to the Lumbardhi; informs proposed stakeholder consultation and governance priorities.

3. Results and Discussion

3.1 Environmental Quality and Anthropogenic Pressures

Faiku et al. (2015) provide the most direct water-quality evidence considered in this synthesis. Their four sampling sites, identified as Haxhaj, "The city", "Transit bridge", and "Near the bridge of Klina", were sampled in June 2013. The reported measurements therefore describe that sampling period and do not represent current

water quality. Across the four sites, pH ranged from 8.20 to 8.58, electrical conductivity from 216 to 399 μS cm^{-1}, nitrate from 0.01 to 1.10 mg dm^{-3}, and nitrite from 0.015 to 0.045 mg dm^{-3}. The study also analyzed 67 elements and reported spatial variation in some concentrations (Faiku et al., 2015). Dinaj (2021) provides complementary evidence from four study areas, covering physicochemical, macropollutant, and microbiological parameters. Management decisions therefore require updated monitoring with clearly identified sampling locations. Figure 1 provides visual context for the urban river setting.



Figure 1. Interaction between the Lumbardhi River and the built environment in the urban segment of Peja
Note: Photograph by the author B.K., October 2026.

Modification of the riverbanks can improve access but may also constrain riparian vegetation and natural river processes. Guimarães et al. (2025) emphasize the need to consider environmental, social, and economic dimensions together in urban river restoration. Water quality is also part of destination quality: visitors experience environmental conditions through the appearance, odor, cleanliness, perceived safety, and attractiveness of riverfront spaces. Walking, cycling, relaxation, and nature observation therefore depend on environmental conditions that support both ecosystem functions and visitor use.

Tourism can itself create pressures when food services, accommodation, events, or increased public use generate poorly managed wastewater, litter, or disturbance. Dinaj (2021) is relevant because the study considers pressures associated with industrial and food-service activities along the Lumbardhi. Sustainable tourism planning consequently requires effective environmental services, enforcement, waste control, wastewater management, and regular monitoring alongside visitor facilities.

A practical monitoring framework could combine ecological indicators with indicators of visitor use and site condition. Ecological monitoring could include repeated physicochemical and, where appropriate, microbiological measurements, together with observations of riparian vegetation and bank disturbance. Monitoring related to recreation could record litter, pedestrian-area condition, unauthorized access, visual degradation, maintenance needs, and conflicts with sensitive habitats. These observations could help the municipality and tourism stakeholders determine when maintenance, access design, or protective measures need to change. This integrated approach is consistent with restoration literature that considers environmental, social, and economic dimensions together (Guimarães et al., 2025).

3.2 Recreational and Tourism Potential

The Lumbardhi River's location within Peja offers opportunities for walking, cycling, leisure, and other recreational activities. These opportunities depend on accessibility, landscape and ecological quality, vegetation, and the arrangement of public spaces. Research on urban riverfronts shows that access and public-space organization influence recreational interaction between cities and rivers (Durán Vian et al., 2021). Applying these

insights to Peja requires attention to local variations in urbanization, bank modification, vegetation, and existing use.

Recreational provision should therefore be differentiated. Central urban segments may offer opportunities for frequent pedestrian use and short periods of leisure, subject to environmental and flood-risk assessment. Sections toward Rugova should retain a stronger natural character and lower-intensity access. This approach is consistent with research emphasizing the importance of natural attractions and their responsible use in the Peja region (Kuqi & Vishaj, 2024). Figure 2 illustrates the relationship between the river and adjacent public and pedestrian spaces in Peja.



Figure 2. Integration of the Lumbardhi River with public, pedestrian, and recreational spaces in Peja
Note: Photograph by the author B.K., May 2026.

Research on Peja's sustainable tourism potential and Rugova's cultural heritage suggests that natural and cultural resources can form complementary parts of the visitor experience (Kuqi & Vishaj, 2024; Lajçi et al., 2022). A survey of 52 visitors in the Rugova Valley found high satisfaction with their experience in the Accursed Mountains National Park (Krasniqi & Morina, 2023). The Lumbardhi could connect urban public space with mountain tourism through a sequence of low-impact experiences. In the urban section, possible activities include walking, relaxation, interpretation of the river and city landscape, and connections to existing commercial and cultural areas. Where conditions permit, cycling routes could complement pedestrian movement without compromising pedestrian safety or requiring unnecessary engineering of the banks. Transition sections could link urban recreation with nature-oriented use, while experiences toward Rugova could emphasize landscape character, vegetation, water, and geological features.

Accessibility also depends on clear information, safety, and comfort. Visitors need to understand route destinations, journey times, terrain, and appropriate behavior in sensitive areas. Durán Vian et al. (2021) identify the importance of public-space location and access to water, while Sousa et al. (2025) describe the cultural and regulating services provided by urban rivers. For Peja, investment should respond to connections among residential areas, visitor routes, public spaces, tourism services, and environmentally suitable access points. Signage, route information, resting places, and interpretation may improve usability with less disturbance than major construction, although site-specific effects still require assessment.

Tourism services can be linked to the corridor without concentrating commercial development along the riverbanks. Guided walks, cycling services, local food, small-scale accommodation, cultural interpretation, and excursions toward Rugova could connect visitors with local businesses. Community-based tourism approaches emphasize local linkages as a means of reducing economic leakage and creating value for residents (Mtapuri et al., 2022). In Peja, the river could organize access to locally supplied experiences while limiting pressure for continuous construction beside the water.

3.3 Hydrological Risk and Ecological Continuity

The study corridor comprises the Lumbardhi river space within urban Peja and its westward, upstream continuation toward Rugova, where the river can link urban public space with nature-oriented and mountain tourism. The analysis does not treat the full hydrological catchment, the river's entire downstream course, or the downstream wastewater-treatment-plant/lake corridor as equivalent to this study area. This distinction is important because the Cityförster concept discussed below primarily extends from the city toward the wastewater treatment plant and a proposed lake, whereas the present manuscript emphasizes the city-to-Rugova direction.

The supporting evidence applies to the study corridor at different spatial scales. Faiku et al. (2015) report four historical sampling sites—Haxhaj, "The city", "Transit bridge", and "Near the bridge of Klina"—but the published labels do not provide a uniform georeferenced basis for assigning all four sites to the Peja–Rugova-oriented corridor; they are therefore treated as historical local/regional evidence rather than a continuous corridor profile. Dinaj (2021) describes four study areas along the Lumbardhi i Pejës and is used to identify types of anthropogenic pressure, without inferring current segment conditions. Agaj et al. (2024) operate at catchment scale and therefore inform flood-risk sensitivity rather than the condition of any individual tourism segment. Locations or boundaries that cannot be established reliably from the cited sources remain explicitly unresolved.

Hydrological risk constrains development along the corridor. Agaj et al. (2024) showed that urbanization influences flood extent and dynamics in the Lumbardhi i Pejës and Klina catchments and emphasized the importance of high-resolution spatial data for resilient planning. Paths, rest areas, and other recreational infrastructure should therefore be assessed against flood exposure as well as accessibility and landscape appeal. Preserving permeable surfaces, riparian vegetation, and space for river processes may support both hydrological resilience and recreational quality. Figure 3 provides visual context for the corridor toward Rugova; it does not constitute a systematic habitat assessment.



Figure 3. Natural characteristics, vegetation, and landscape of the Lumbardhi corridor toward Rugova
Note: Photograph by B.K., October 2026.

The available sources do not provide a standardized, current comparison of sites throughout the corridor from Peja toward Rugova. The proposed differentiation is therefore a preliminary planning interpretation. The photographs provide visual context but do not constitute a systematic field inventory.

Preliminary assessment could consider the degree of urbanization and interaction with the built environment; documented water-quality or pollution concerns; riparian vegetation and ecological continuity; flood exposure and bank stability; existing access and recreational use; and proximity to established tourism services. These criteria organize the available information but do not yield validated segment scores or boundaries. Research on river restoration and riverfront recreation supports consideration of the relationships among access, infrastructure, ecosystem services, and recreational use (Durán Vian et al., 2021; Guimarães et al., 2025; Kaiser et al., 2020; Sousa et al., 2025). It does not establish the effects of particular interventions along the Lumbardhi. Any proposed urban-to-Rugova differentiation should therefore be tested through updated flood mapping, bank and habitat assessments, and monitoring of actual use before site-specific decisions are made.

3.4 Implications for Sustainable Tourism Development

The synthesis suggests that the Lumbardhi's tourism potential depends on more than visual appeal. Water quality, riverbank condition, vegetation, accessibility, flood safety, and continuity between Peja and Rugova are interdependent. Urban rivers provide regulating and cultural ecosystem services (Sousa et al., 2025), and restoration can improve some services (Kaiser et al., 2020).

Potential urban priorities include pollution control, ecological rehabilitation, pedestrian connections, and selective improvements to access. Toward Rugova, assessment should emphasize riparian vegetation, landscape character, and ecological continuity. The appropriate intensity of use must be established through further environmental and risk assessment; it cannot be inferred from the availability of space for infrastructure.

Cityförster (2023) presents the 'Revitalizing Peja's River—Kosovo's Green Corridor' concept, prepared as technical assistance to the Municipality of Peja for the Netherlands Enterprise Agency (RVO). The project page lists the concept at the idea stage for 2022–2023. The documented concept covers about 52 ha and is organized around four design components: a City Green Spine, River Park, Natural Wetlands, and a Multi-purpose Lake. Its stated spatial sequence runs from the city centre downstream toward the wastewater treatment plant and lake, with walking/cycling connections and a proposed shuttle. The present synthesis focuses primarily on the upstream direction from urban Peja toward Rugova. The two approaches overlap in the urban river section and cover different stretches beyond it. The Cityförster project presents a design and pre-feasibility concept. Its implementation was outside the scope of this synthesis, which organizes published evidence and its limitations around environmental quality, riparian condition, flood exposure, recreation, and local benefits.

Historical water-quality observations describe conditions at the time of sampling, catchment-scale flood research informs risk screening, and improvements to access remain conditional on habitat and risk assessment. The framework can help assess whether and where interventions of the kind proposed by Cityförster would be environmentally and socially appropriate. These questions require site-specific assessment.

The corridor's benefits should also extend to residents and local enterprises. Participation, local ownership, and the distribution of benefits are central concerns in sustainable tourism (Blackstock, 2005; Chambwe & Saayman, 2025; Simpson, 2009). River-related experiences could connect visitors with guides, hospitality businesses, food services, small enterprises, and cultural activities already present in Peja and Rugova. Such connections should be developed without excessive commercialization of the riverbanks.

The corridor's contribution could be assessed through four linked outcomes: environmental integrity, visitor value, local economic linkages, and community legitimacy. Environmental assessment should consider water quality, riparian vegetation, and ecological continuity. Visitor assessment should consider safe, clear, and attractive routes and public spaces. Economic assessment should examine opportunities for locally owned services and enterprises, while community assessment should examine residents' influence over decisions and the distribution of costs and benefits. These outcomes are interdependent: ecological improvement alone does not guarantee local economic benefits, and tourism investment without meaningful participation may generate conflict.

Studies of Kosovo provide context for connecting these outcomes. Kuqi & Vishaj (2024) discuss Peja's natural attractions, cultural heritage, and hospitality; Lajçi et al. (2022) examine Rugova's cultural heritage. Together, these studies support considering the Lumbardhi within a broader system of public spaces, river landscapes, cultural resources, and mountain tourism.

The wider Kosovo literature also supports attention to local landscape and culture. Tahiri et al. (2020) discuss sustainable tourism development in Kosovo, and Tahiri et al. (2022) connect tourism with regional heritage, local food, traditions, and diversity. Kuqi (2025) focuses on mountain tourism in the Rugova area of Peja. These sources provide context for exploring how the Lumbardhi corridor could connect urban experiences with local culture, food, and mountain activities, subject to ecological protection and site-specific assessment.

3.5 Community Participation, Local Benefits, and Tourism Governance

Community participation is most useful when residents can influence specific decisions and share in the resulting benefits. Blackstock (2005) cautions against treating participation as symbolic, while Choi & Murray (2010) and Choi & Sirakaya (2005) provide frameworks for understanding residents' attitudes toward sustainable tourism. For the Lumbardhi, these perspectives support consultation before major interventions, clear communication of environmental and tourism objectives, and opportunities to influence access, public-space design, events, and visitor services. They provide concepts for local assessment rather than a destination model to be transferred directly to Peja.

Benefit distribution also requires explicit attention. Saputra et al. (2025) consider economic, sociocultural, and environmental dimensions of community well-being. These perspectives raise practical questions for Peja: whether investment improves public space, whether local enterprises can participate in tourism value chains, whether environmental costs are controlled, and whether residents retain access to the corridor. Local indicators and periodic community surveys could help address these questions.

A practical governance model should coordinate environmental monitoring, tourism planning, and local economic development. Literature on community-based tourism emphasizes participation, livelihoods, and residents' perceptions (Lee & Jan, 2019; Mtapuri et al., 2022; Simpson, 2009). Updated water-quality information, flood mapping, visitor-use data, and resident feedback should inform decisions about access, events, commercial activity, and rehabilitation. Participation should begin during problem definition and continue through route selection, identification of sensitive sites, and assessment of business opportunities. Residents can contribute knowledge of seasonal crowding, informal access, waste, safety, and riverbank changes; businesses can identify visitor needs; and environmental specialists can assess ecological constraints. These forms of knowledge can support decisions that account for both technical requirements and community priorities.

The main stakeholder groups affected by corridor decisions include residents using riverfront public space; residents and landholders near access or restoration sites; hospitality, food-service, guiding, cycling, accommodation, and other tourism enterprises; municipal tourism and planning staff; environmental and infrastructure services; and civil-society organizations.

Potential benefits include improved public space, safer and clearer access, visitor expenditure, and opportunities for locally supplied services. Potential costs include disturbance, waste, restricted everyday access, flood exposure, ecological degradation, or unequal capture of tourism income. Participation should therefore be tied to specific decisions—route alignment, event locations, restoration priorities, access restrictions, and business opportunities—with responsibilities and unresolved conflicts recorded rather than treating 'the community' as a single interest group.

Peja's Tourism Sector and Tourist Information Office could support corridor planning, visitor information, and stakeholder consultation. Before implementation, the municipality and relevant public-service bodies should clarify who will be responsible for wastewater and solid waste management, public-space maintenance, riverbank works, and infrastructure exposed to flooding. Procurement and tourism-product development could strengthen local benefits. Destination managers could examine the share of services supplied by local enterprises and the range of residents able to participate, alongside total visitor expenditure. Community-based tourism approaches emphasize local linkages and reduced economic leakage (Mtapuri et al., 2022). Opportunities might include guided walks or cycling, food and craft products associated with Peja and Rugova, maintenance and interpretation services, accommodation packages, and cooperation between urban businesses and rural providers. Their viability would depend on demonstrated demand and environmental capacity.

A transparent indicator system could make this approach operational. Environmental indicators could include repeated water-quality measurements, visible waste, riparian vegetation condition, and the extent of bank degradation. Recreational indicators could include route continuity, maintenance, observed use, visitor satisfaction, and perceived safety. Community indicators could track participation in planning, locally owned businesses, local employment, resident perceptions, and conflicts with everyday river use. Hydrological assessment should consider infrastructure exposure and consistency with flood-resilience planning. The purpose would be to reveal trade-offs rather than generate a single sustainability score. Deteriorating environmental conditions alongside increased use would indicate a need to adjust management; weak participation despite environmental improvement would indicate a need to strengthen governance and local economic links.

Implementation could proceed in phases. The first phase would establish a baseline through environmental monitoring, geographic information system (GIS) mapping of access and flood risk, an inventory of tourism and public-space assets, and consultation with residents and businesses. The second would prioritize low-impact measures in locations where ecological conditions and flood risk permit, such as maintenance, signage, route connections, interpretation, and selective restoration. The third would evaluate observed effects before further infrastructure is authorized. This sequence would allow management to respond to environmental change and tourism demand. Evidence that residents' perceptions vary across stages of tourism development also supports periodic reassessment of governance and management priorities (Lee & Jan, 2019).

3.6 Integrated Management Framework and Research Priorities

This study proposes three preliminary planning categories for the Lumbardhi corridor: an urban recreation area, a transition area, and a nature-oriented area toward Rugova. They are analytical concepts, not mapped or verified zones, and the study does not establish their boundaries or ecological carrying capacities. The proposed differentiation draws on five criteria: urbanization and connections to public spaces; available water-quality and pollution evidence; riparian vegetation and ecological continuity; flood exposure and bank stability; and the type and intensity of recreational use.

When criteria point in different directions, ecological and safety constraints operate as limiting conditions rather than variables that can simply be offset by tourism demand. For example, strong visitor appeal or business interest would not justify intensified access where flood exposure, bank instability, habitat sensitivity, or unresolved pollution indicates unacceptable risk. Where ecological constraints are manageable but community costs or access conflicts are substantial, the preferred response would be redesign, lower-intensity use, or delayed implementation

pending consultation. This qualitative hierarchy makes the framework precautionary: first screen for ecological and hydrological constraints, then assess access and recreational suitability, and finally evaluate local benefit, responsibilities, and monitoring needs. The hierarchy itself remains a proposition to be tested through updated field, spatial, and community evidence.

The categories are differentiated by management emphasis. An urban recreation area would prioritize safe public-space connection, maintenance, pollution control, and low-impact access, but only where flood and habitat conditions permit. A transition area would prioritize continuity between built and more natural settings, selective restoration, and moderated access. A nature-oriented area toward Rugova would place ecological continuity, habitat protection, and controlled access ahead of infrastructure intensity. These relationships are partly source-supported and partly authors' propositions: the literature supports the relevance of water quality, flood risk, river–city access, restoration, ecosystem services, and participation, whereas the specific combination into three Lumbardhi categories is the authors' preliminary synthesis and requires field testing.

The framework links four levels of reasoning. First, the analytical dimensions identify the kinds of evidence to be considered: environmental quality, anthropogenic pressure, recreation/tourism function, and hydrological risk, with governance and local benefit treated across them. Second, the assessment criteria translate those dimensions into place-sensitive questions concerning urbanization/public-space connection, water-quality or pollution evidence, riparian continuity, flood exposure/bank stability, and existing or proposed recreational intensity. Third, the three planning categories are qualitative responses to different combinations of these criteria rather than fixed zones. Fourth, outcomes—environmental integrity, visitor value, local economic linkage, and community legitimacy—provide the basis for evaluating whether a chosen intervention remains acceptable over time.

The evidence varies in date, spatial coverage, and purpose. The June 2013 water-quality campaign, the four-area assessment by Dinaj (2021), and the catchment-scale flood analysis by Agaj et al. (2024) provide different kinds of information. Together, they inform further investigation but do not validate a spatial classification of the corridor.

Establishing boundaries and appropriate levels of use requires updated water-quality sampling, field assessments of riparian habitats and riverbanks, GIS-based mapping of land use and flood exposure, and systematic monitoring of recreation. Institutional coordination and community participation should guide these assessments and subsequent decisions. The framework provides a preliminary basis for integrating ecological protection, recreation, and tourism planning; its practical application depends on further field evidence and spatial analysis.

4. Conclusions

The Lumbardhi River is an ecological, landscape, and recreational asset for Peja with potential to contribute to sustainable tourism, provided that ecological and hydrological constraints are addressed. The reviewed sources document historical water-quality observations from June 2013, anthropogenic pressures, and catchment-scale flood risk. They do not establish current conditions throughout the corridor or the ecological carrying capacity of specific locations. The proposed differentiation from urban Peja toward Rugova should therefore be treated as a preliminary planning framework, not a verified classification or mapped zoning result.

Management priorities include pollution prevention, updated environmental monitoring, protection and restoration of riparian vegetation, and access planning informed by flood risk and habitat condition. The available evidence does not demonstrate that any specific segment can accommodate increased visitor use without ecological harm.

Further assessment should combine seasonal water-quality monitoring, field surveys of habitats and riverbanks, GIS-based analysis of land use and flood exposure, and evaluation of recreational impacts and community priorities. The framework can organize these assessments and support adaptive management. Meaningful local participation and fair distribution of tourism benefits should accompany environmental safeguards so that future development responds to community needs while protecting the river's ecological integrity.

Author Contributions

Conceptualization, B.K.; methodology, B.K.; formal analysis, B.K.; investigation, B.K. and A.D.; resources, B.K. and A.D.; data curation, B.K.; writing—original draft preparation, B.K.; writing—review and editing, B.K. and A.D.; visualization, B.K.; project administration, B.K. and A.D. Both authors have read and agreed to the submitted version of the manuscript.

Data Availability

No new datasets were generated in this study. The evidence analyzed is available in the published sources cited in the reference list.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

Generative AI and AI-assisted technologies were used for English-language translation, language editing, citation-consistency checking, and manuscript-formatting support. The authors reviewed and revised the manuscript and remain responsible for its content, accuracy, originality, and references.

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